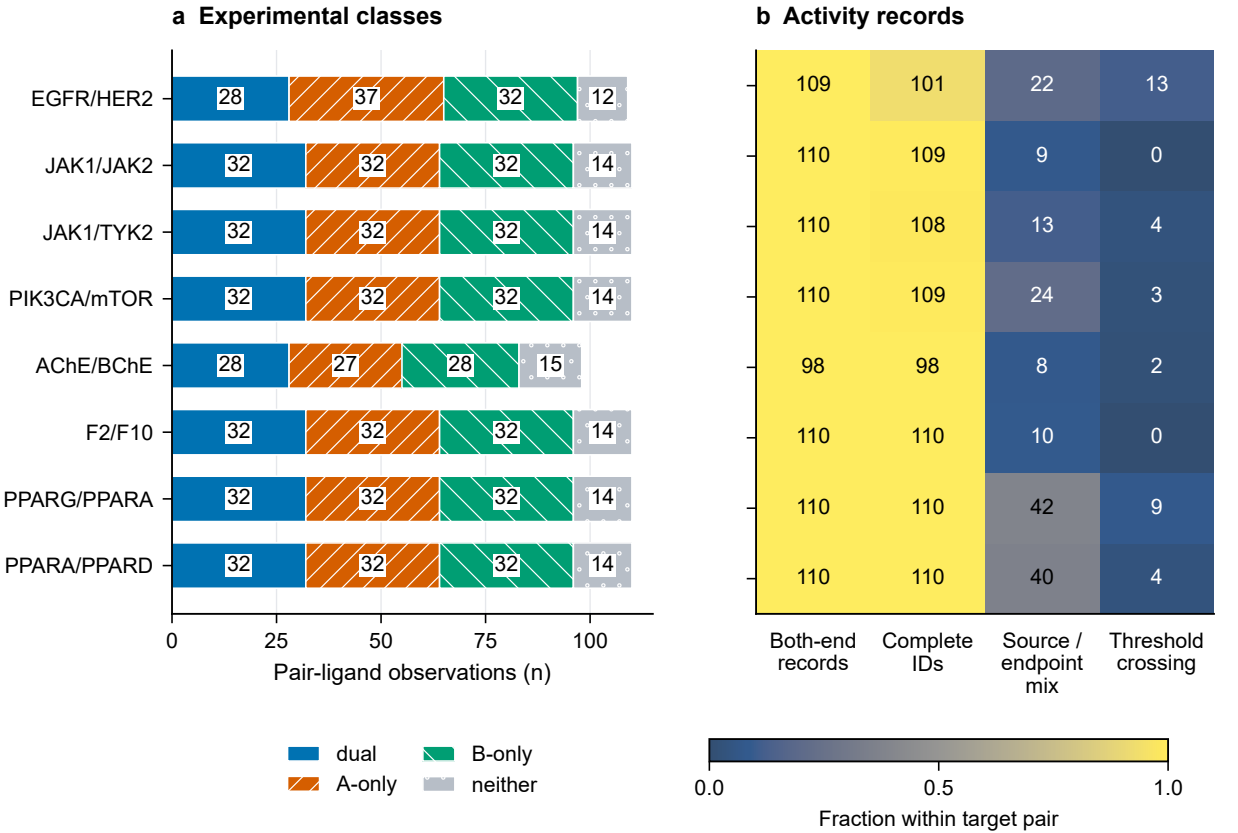
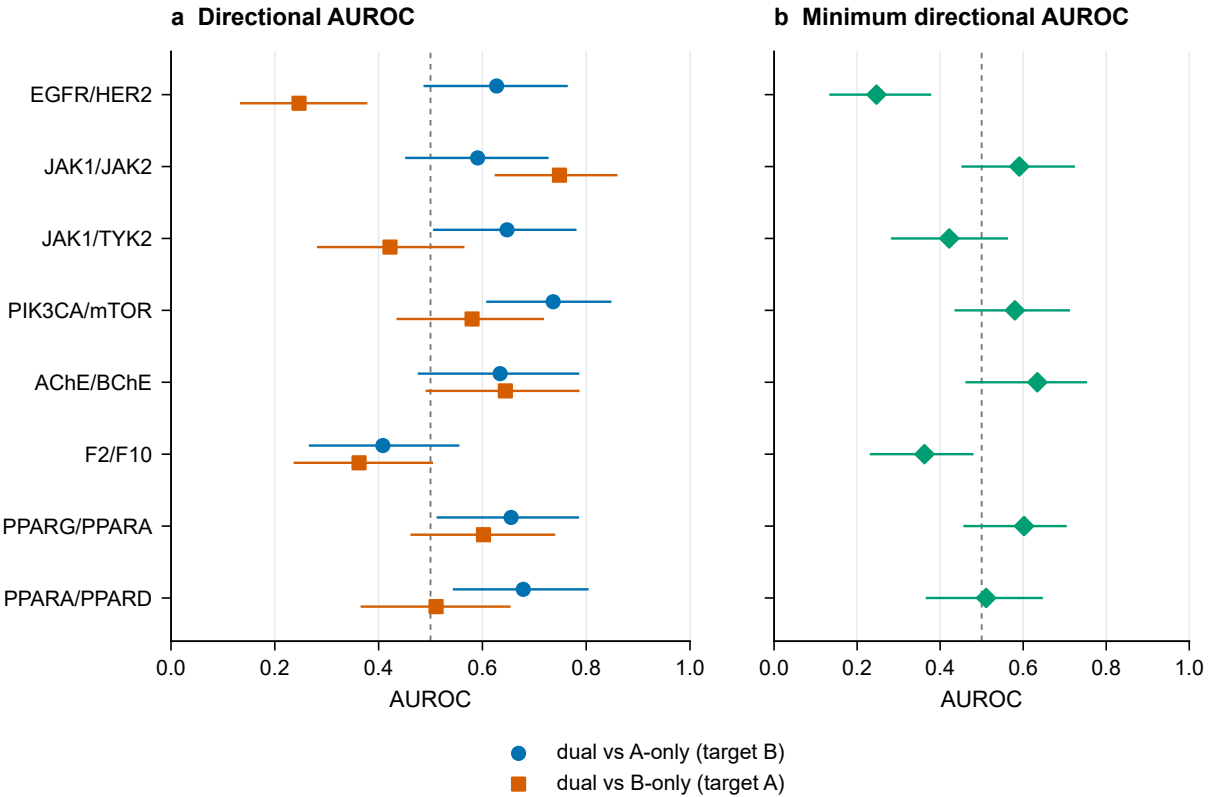
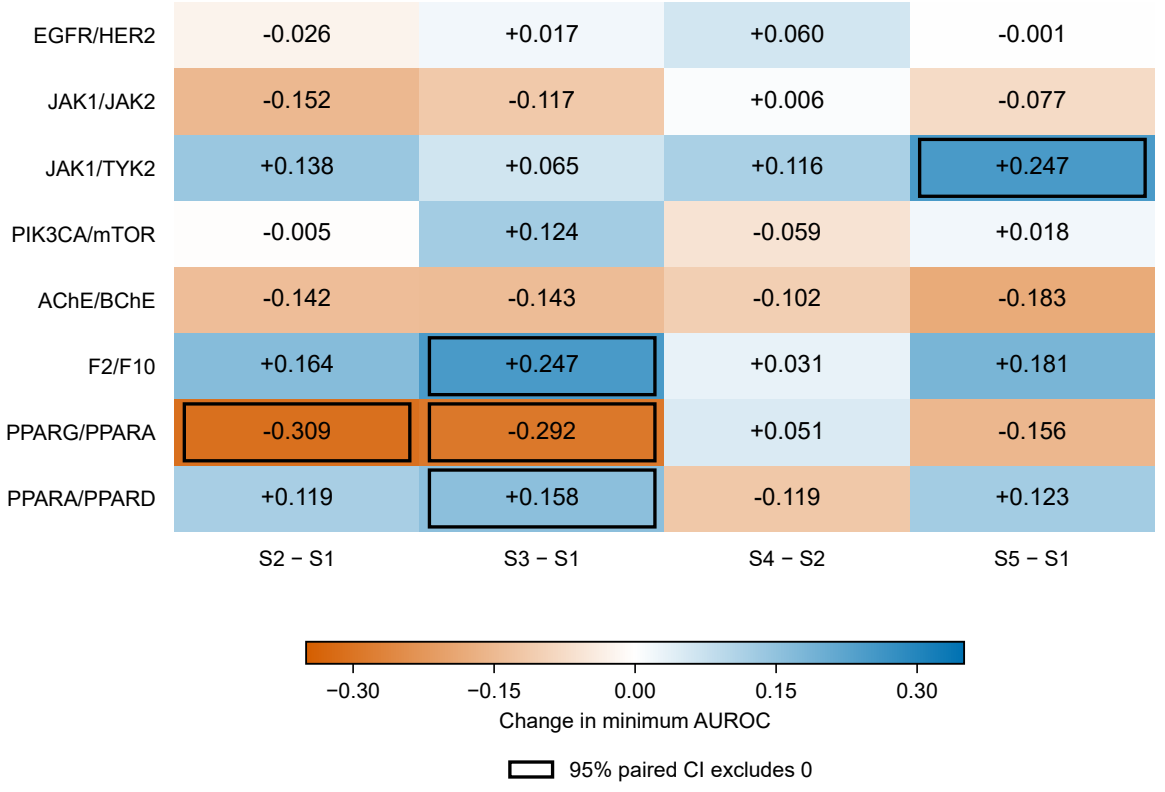
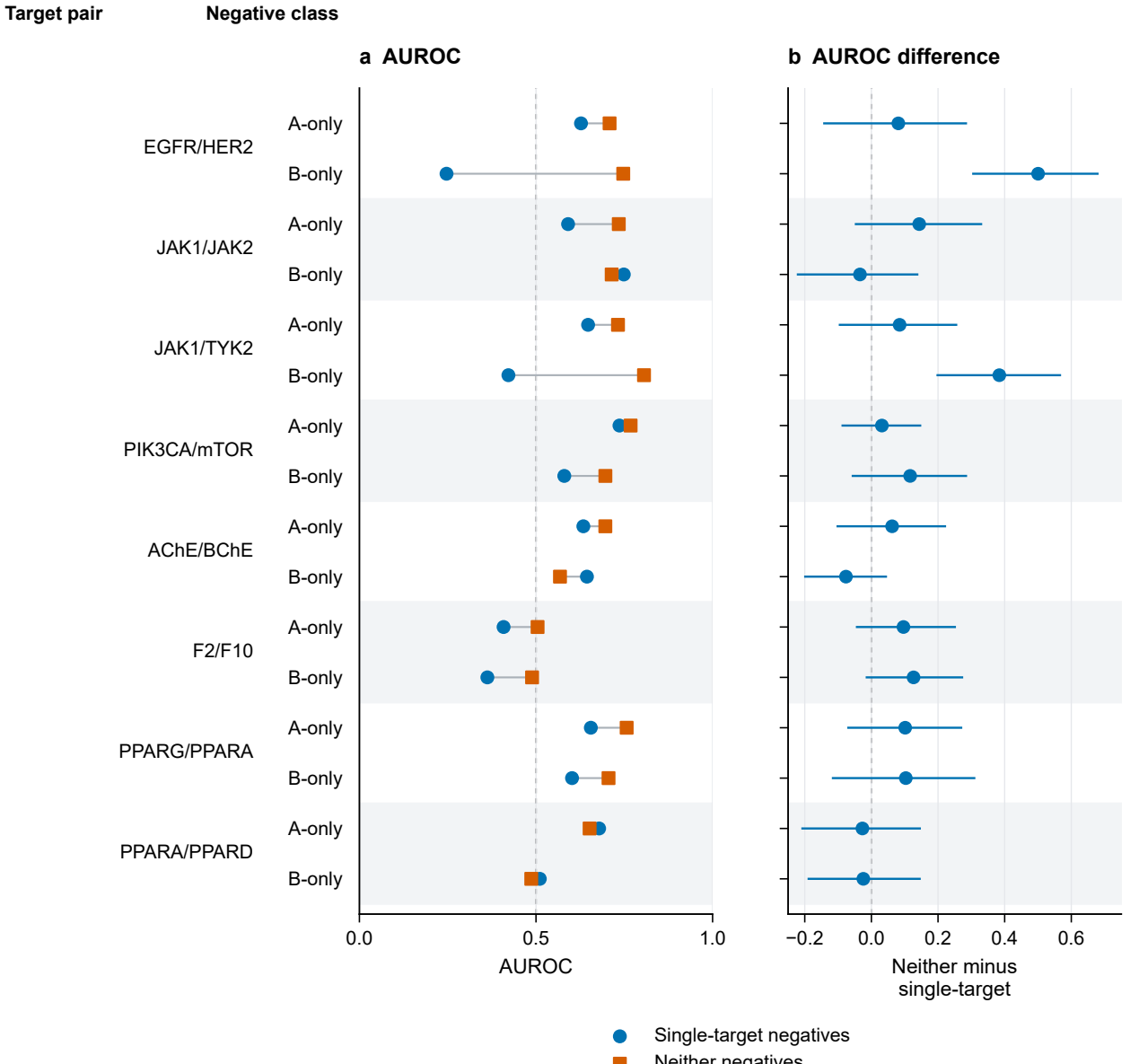
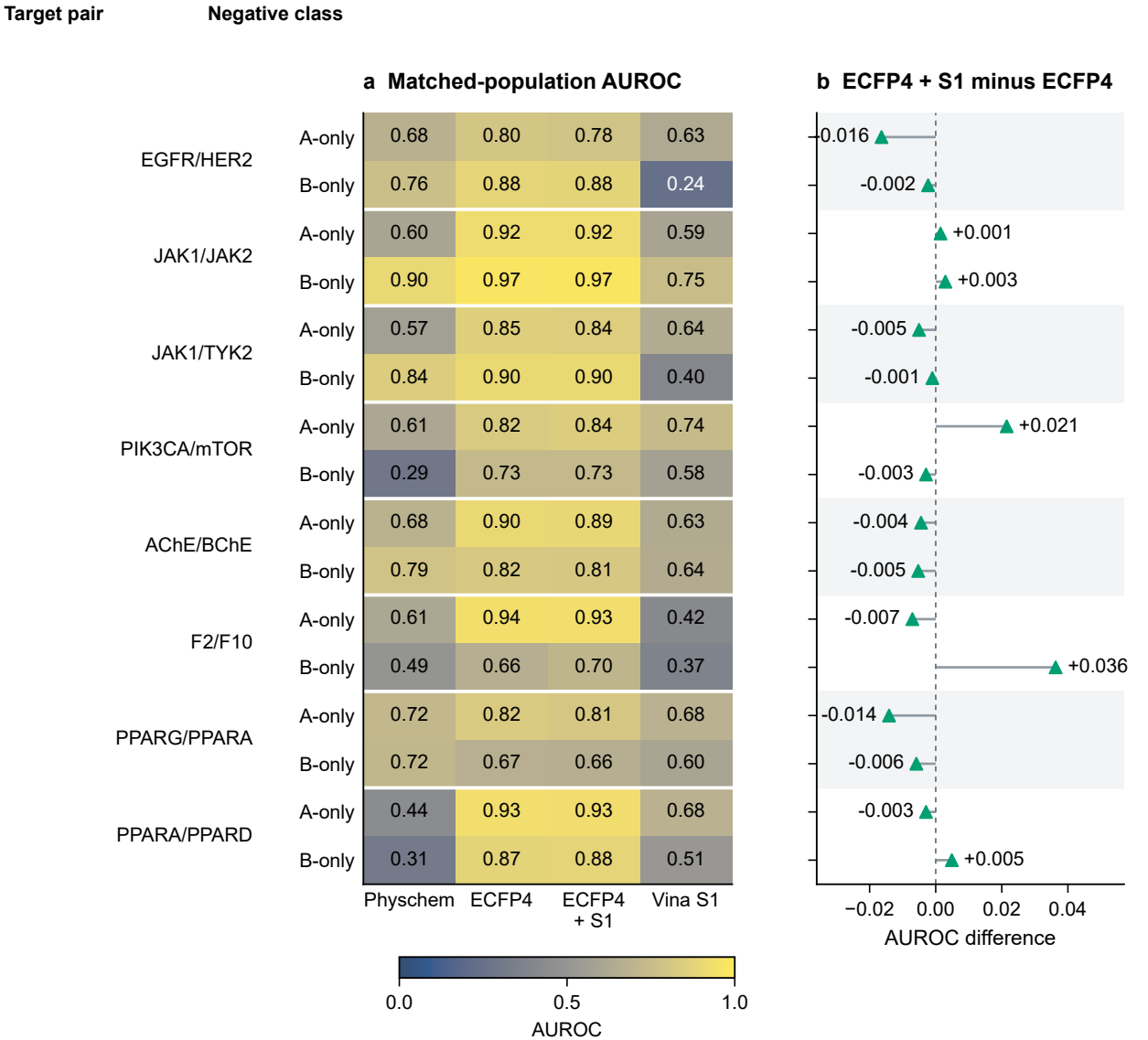
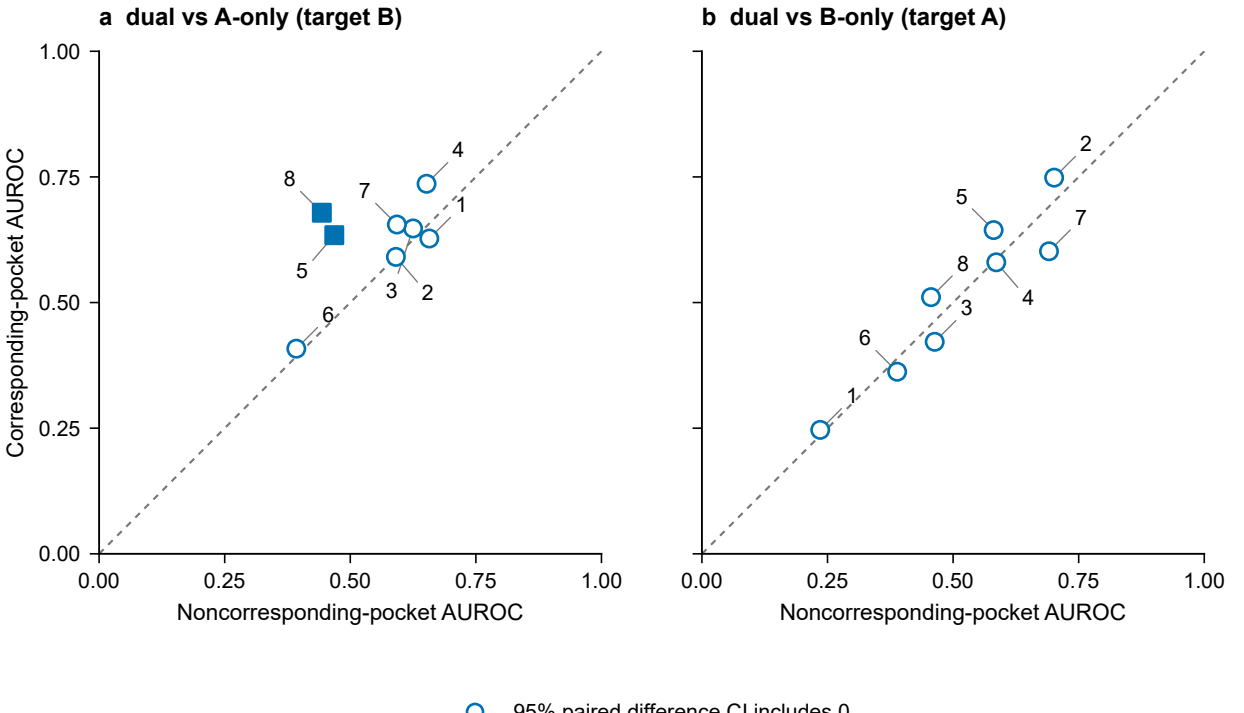

a Experimental classes
EGFR/HER2
JAK1/JAK2
JAK1/TYK2
PIK3CA/mTOR
AChE/BChE
F2/F10
PPARG/PPARA
PPARA/PPARD
28
37
32
12
32
32
32
14
32
32
32
14
32
32
32
14
28
27
28
15
32
32
32
14
32
32
32
14
32
32
32
14
0
25
50
75
100
Pair-ligand observations (n)
dual
A-only
B-only
neither
b Activity records
109
101
22
13
110
109
9
0
110
108
13
4
110
109
24
3
98
98
8
2
110
110
10
0
110
110
42
9
110
110
40
4
Both-end records
Complete IDs
Source / endpoint mix
Threshold crossing
0.0
0.5
1.0
Fraction within target pair

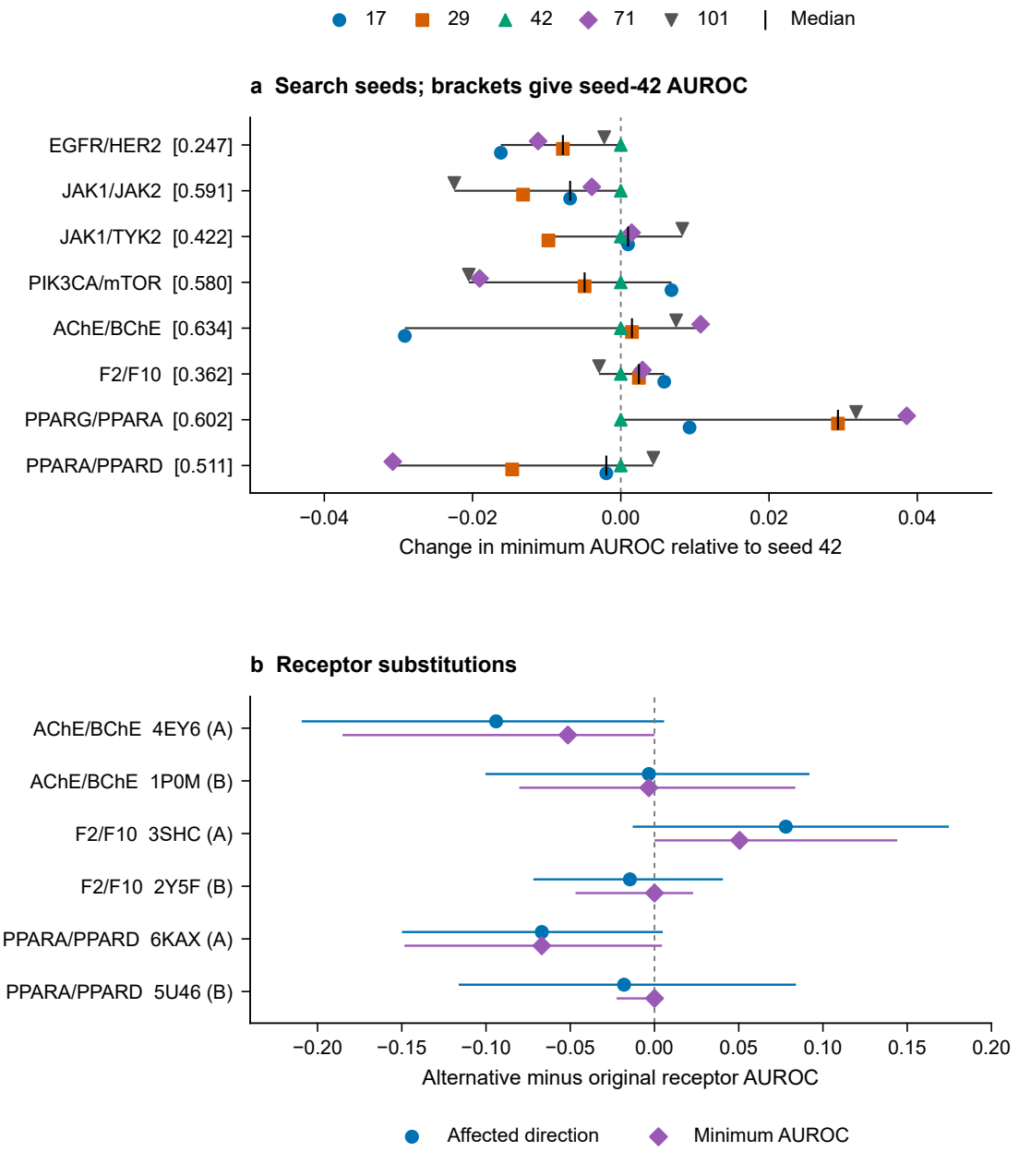

a Directional AUROC
b Minimum directional AUROC
EGFR/HER2
JAK1/JAK2
JAK1/TYK2
PIK3CA/mTOR
AChE/BChE
F2/F10
PPARG/PPARA
PPARA/PPARD
0.0
0.2
0.4
0.6
0.8
1.0
AUROC
dual vs A-only (target B)
dual vs B-only (target A)

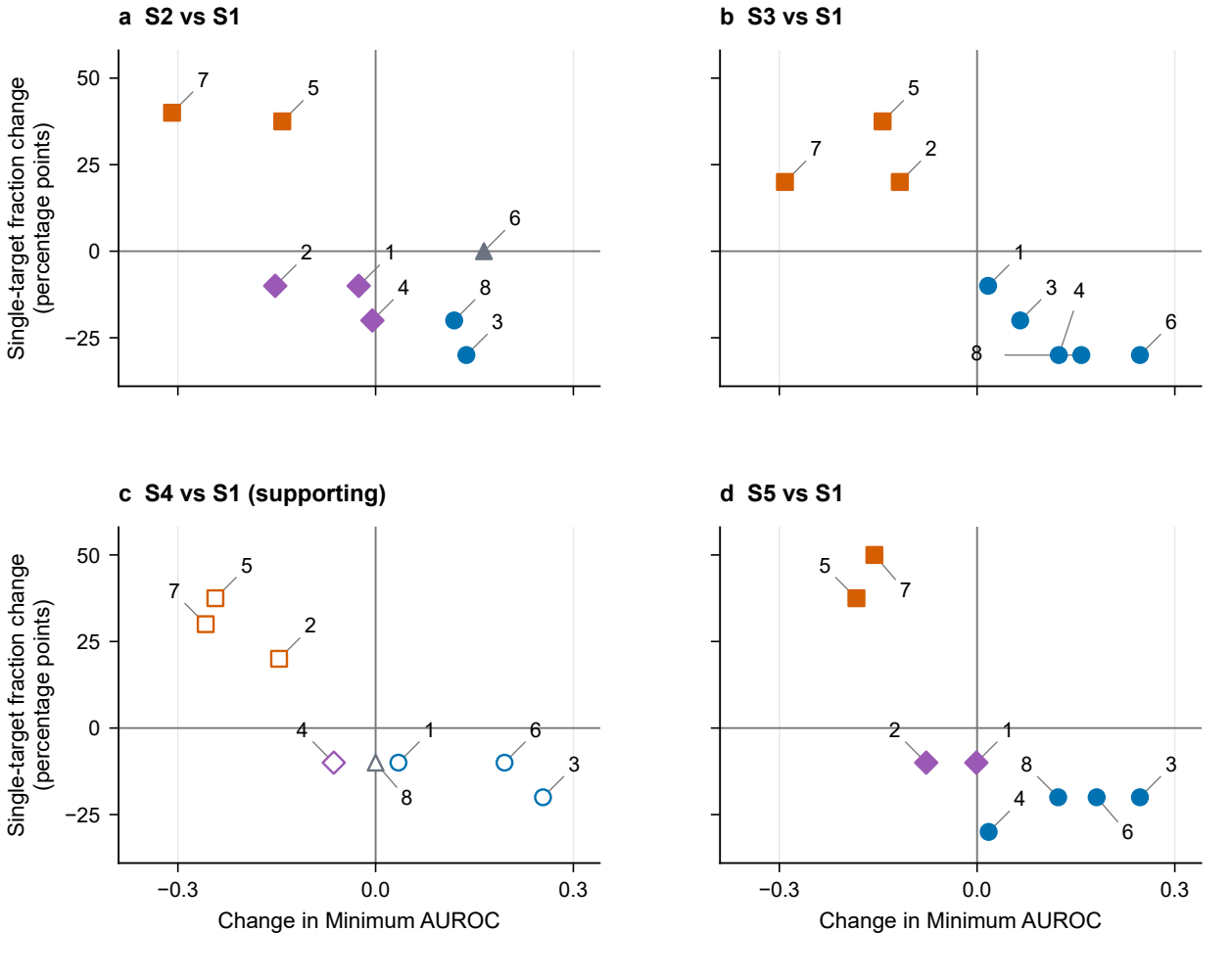

EGFR/HER2
JAK1/JAK2
JAK1/TYK2
PIK3CA/mTOR
AChE/BChE
F2/F10
PPARG/PPARA
PPARA/PPARD
-0.026
+0.017
+0.060
-0.001
-0.152
-0.117
+0.006
-0.077
+0.138
+0.065
+0.116
+0.247
-0.005
+0.124
-0.059
+0.018
-0.142
-0.143
-0.102
-0.183
+0.164
+0.247
+0.031
+0.181
-0.309
-0.292
+0.051
-0.156
+0.119
+0.158
-0.119
+0.123
S2 − S1
S3 − S1
S4 − S2
S5 − S1
−0.30
−0.15
0.00
0.15
0.30
Change in minimum AUROC
95% paired CI excludes 0

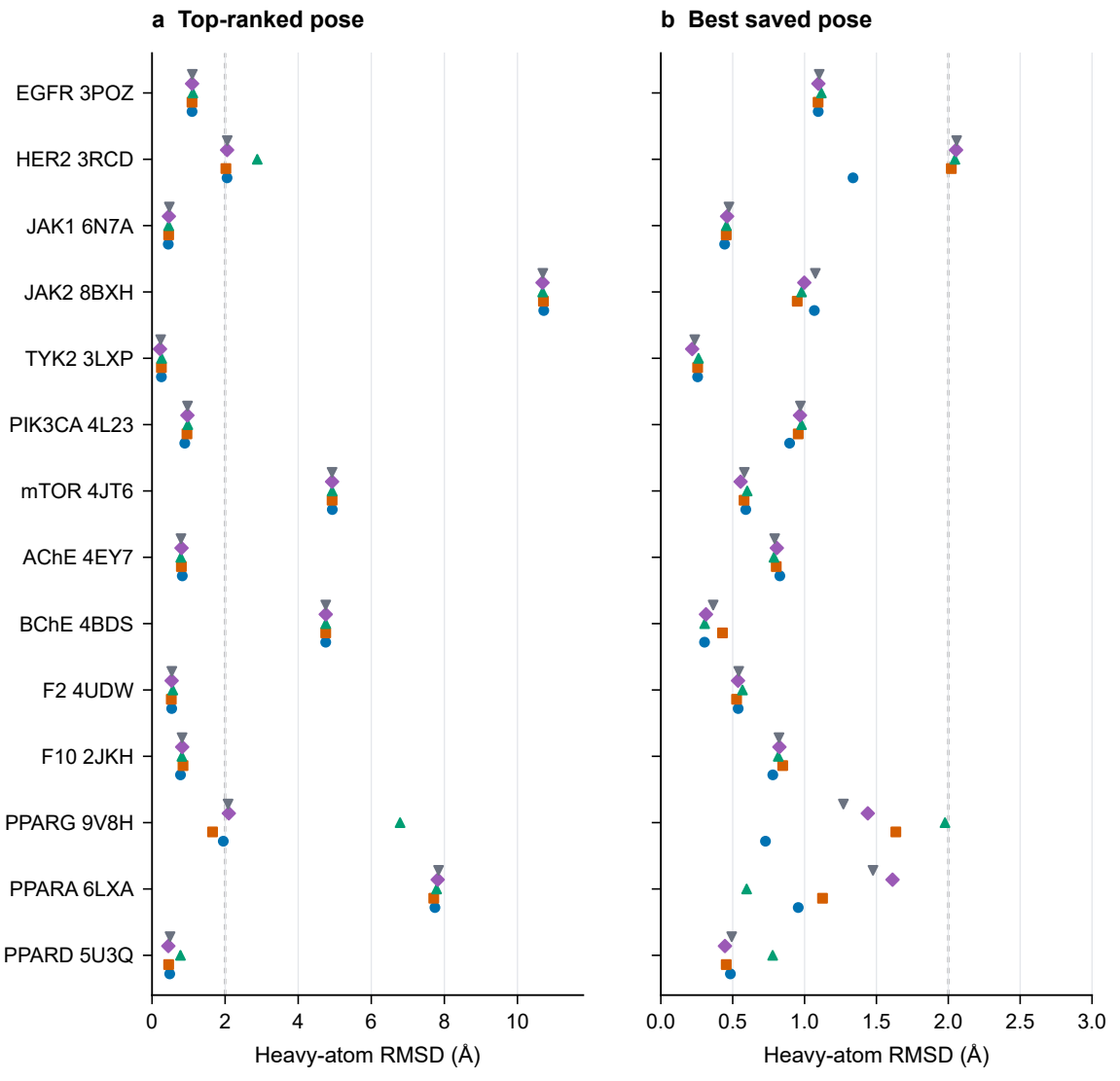

Target pair
Negative class
a AUROC
b AUROC difference
EGFR/HER2
JAK1/JAK2
JAK1/TYK2
PIK3CA/mTOR
AChE/BChE
F2/F10
PPARG/PPARA
PPARA/PPARD
A-only
B-only
0.0
0.5
1.0
AUROC
−0.2
0.0
0.2
0.4
0.6
Neither minus single-target
Single-target negatives
Neither negatives

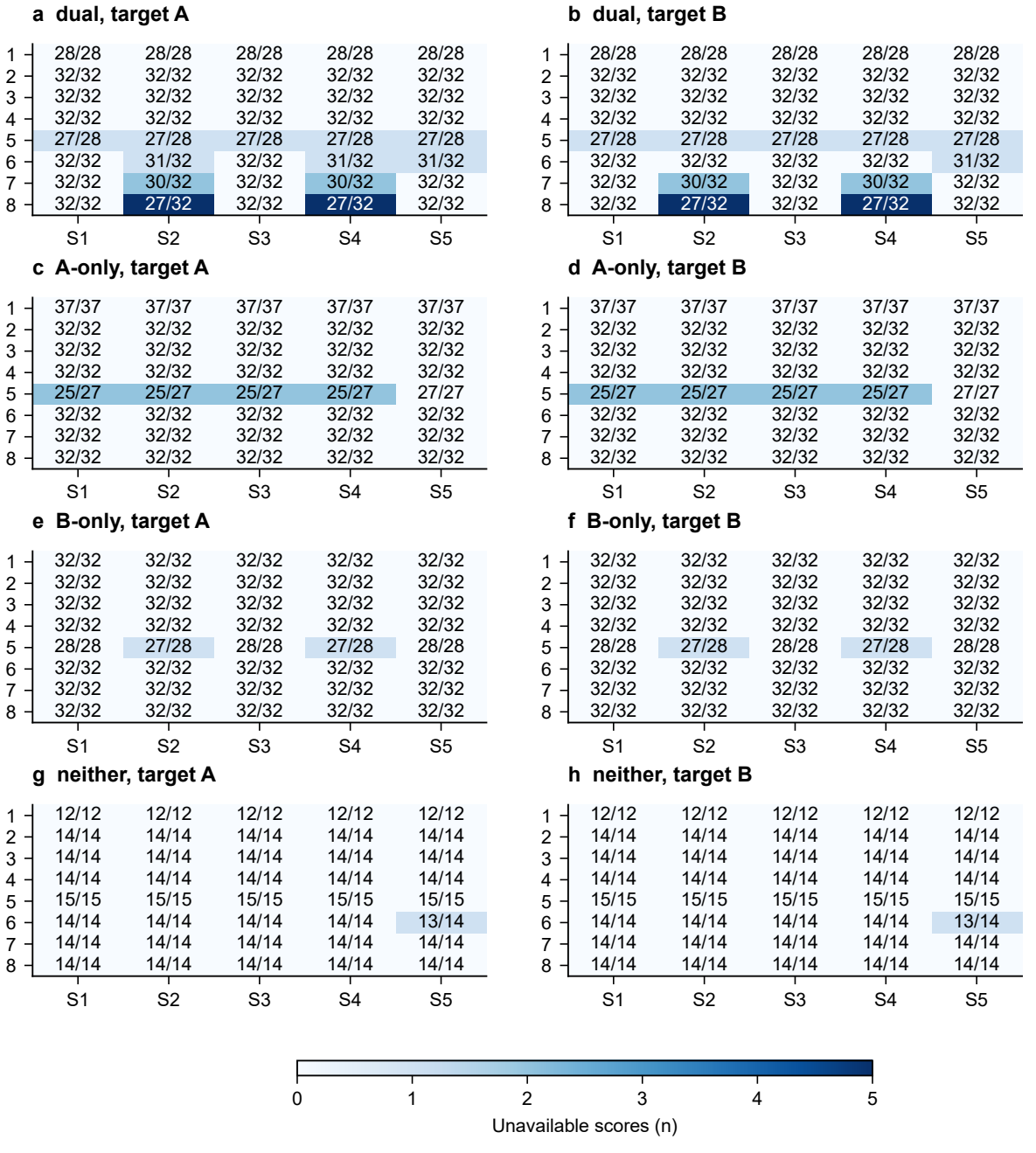

Target pair
Negative class
a Matched-population AUROC
b ECFP4 + S1 minus ECFP4
EGFR/HER2
JAK1/JAK2
JAK1/TYK2
PIK3CA/mTOR
AChE/BChE
F2/F10
PPARG/PPARA
PPARA/PPARD
A-only
B-only
0.68 0.80 0.78 0.63
0.76 0.88 0.88 0.24
0.60 0.92 0.92 0.59
0.90 0.97 0.97 0.75
0.57 0.85 0.84 0.64
0.84 0.90 0.90 0.40
0.61 0.82 0.84 0.74
0.29 0.73 0.73 0.58
0.68 0.90 0.89 0.63
0.79 0.82 0.81 0.64
0.61 0.94 0.93 0.42
0.49 0.66 0.70 0.37
0.72 0.82 0.81 0.68
0.72 0.67 0.66 0.60
0.44 0.93 0.93 0.68
0.31 0.87 0.88 0.51
Physchem
ECFP4
ECFP4 + S1
Vina S1
0.0
0.5
1.0
AUROC
-0.016
-0.002
+0.001
+0.003
-0.005
-0.001
+0.021
-0.003
-0.004
-0.005
-0.007
+0.036
-0.014
-0.006
-0.003
+0.005
−0.02
0.00
0.02
0.04
AUROC difference

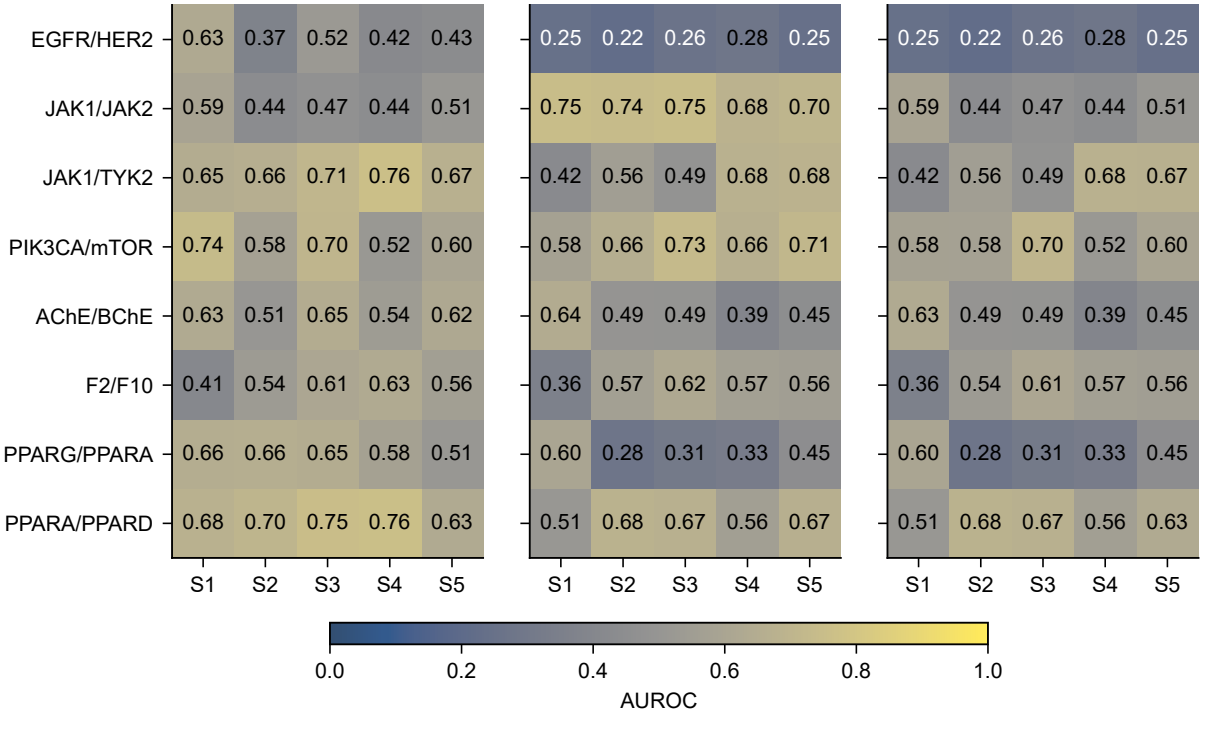

a dual vs A-only (target B)
b dual vs B-only (target A)
Corresponding-pocket AUROC
Noncorresponding-pocket AUROC
0.00
0.25
0.50
0.75
1.00
1
2
3
4
5
6
7
8
95% paired difference CI includes 0
95% paired difference CI excludes 0

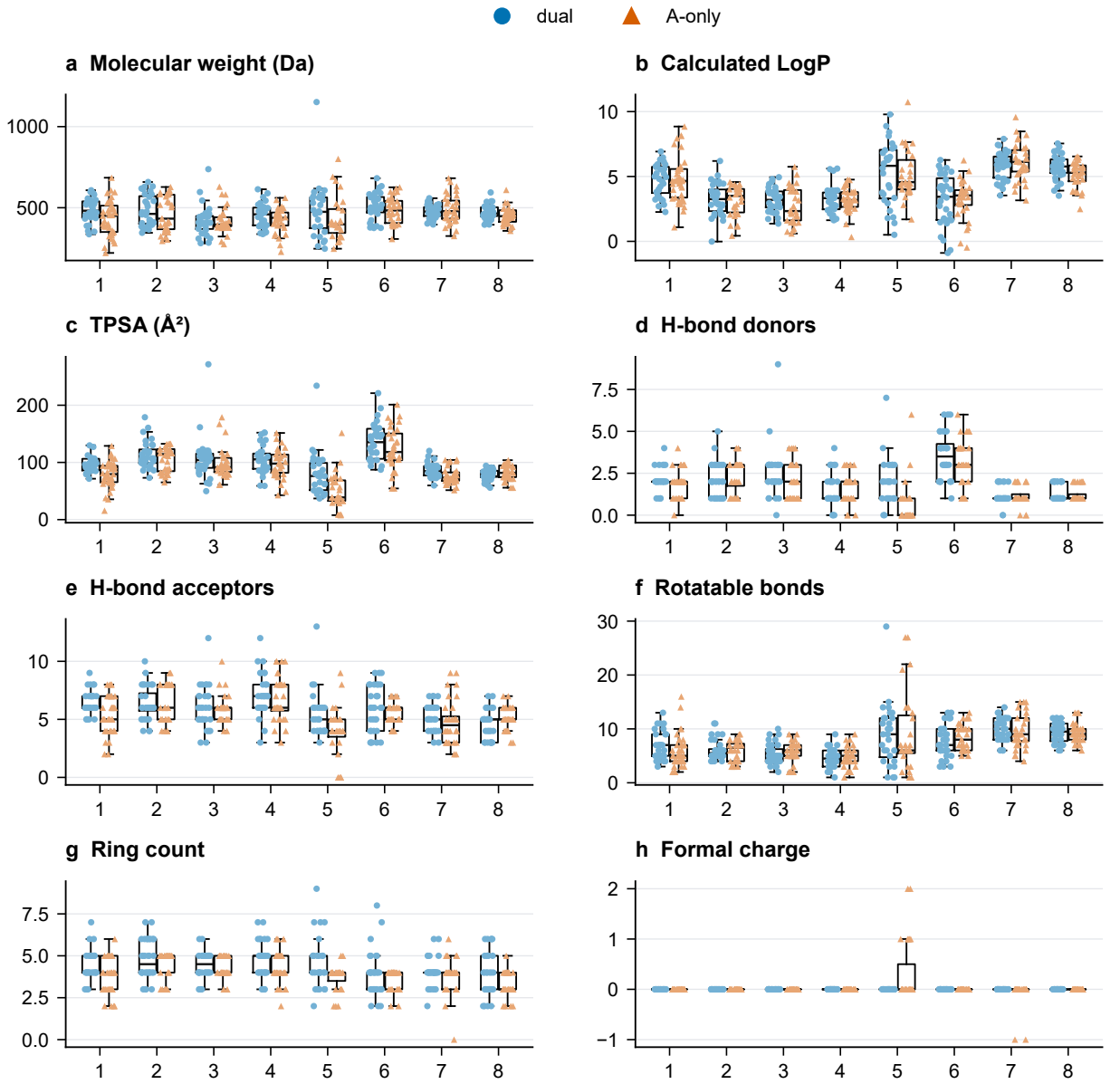

17
29
42
71
101
Median
a Search seeds; brackets give seed-42 AUROC
EGFR/HER2 [0.247]
JAK1/JAK2 [0.591]
JAK1/TYK2 [0.422]
PIK3CA/mTOR [0.580]
AChE/BChE [0.634]
F2/F10 [0.362]
PPARG/PPARA [0.602]
PPARA/PPARD [0.511]
−0.04
−0.02
0.00
0.02
0.04
Change in minimum AUROC relative to seed 42
b Receptor substitutions
AChE/BChE 4EY6 (A)
AChE/BChE 1P0M (B)
F2/F10 3SHC (A)
F2/F10 2Y5F (B)
PPARA/PPARD 6KAX (A)
PPARA/PPARD 5U46 (B)
−0.20
−0.15
−0.10
−0.05
0.00
0.05
0.10
0.15
0.20
Alternative minus original receptor AUROC
Affected direction
Minimum AUROC

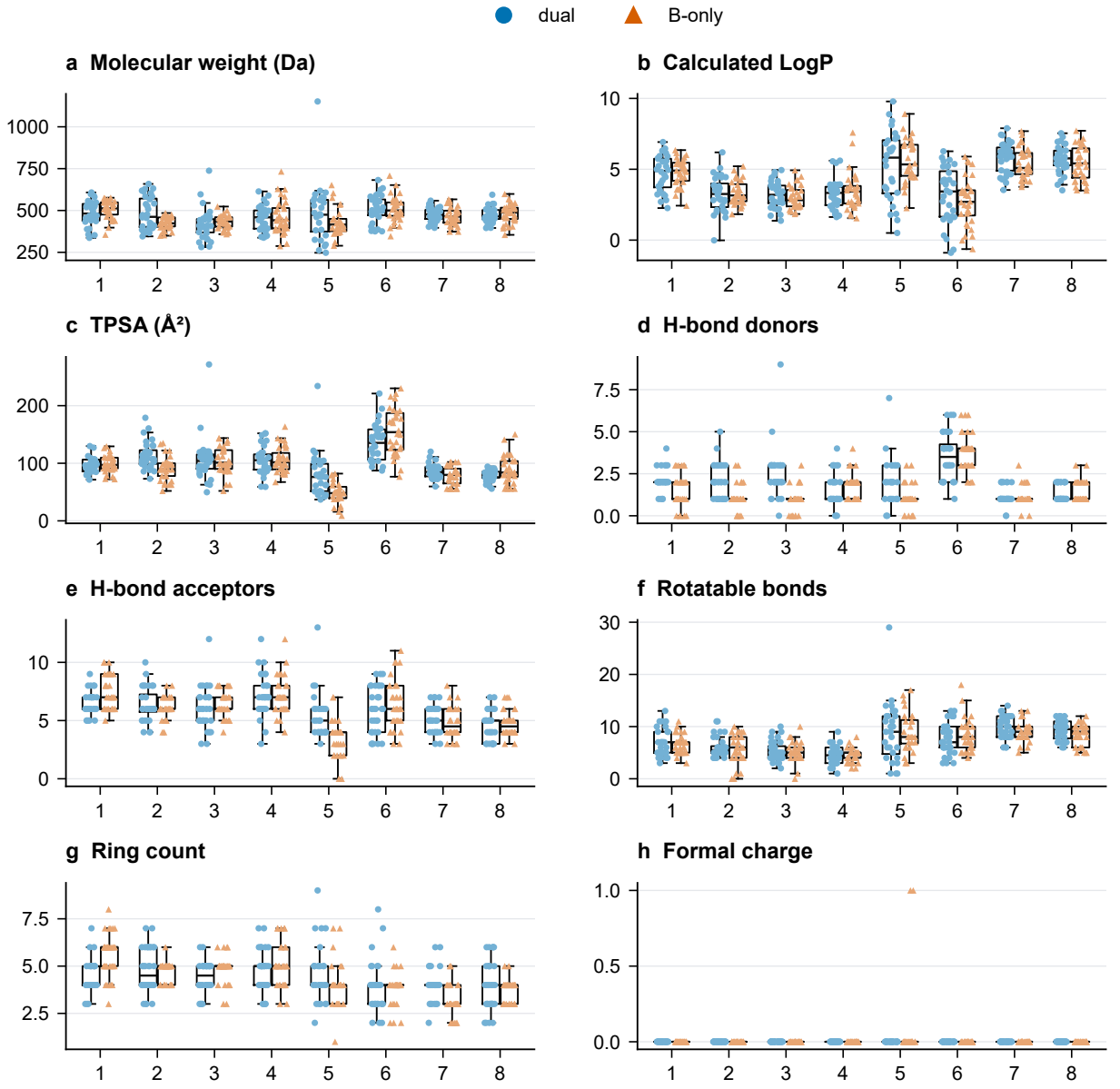

a S2 vs S1
b S3 vs S1
c S4 vs S1 (supporting)
d S5 vs S1
Single-target fraction change (percentage points)
Change in Minimum AUROC
50
25
0
−25
−0.3
0.0
0.3
Concordant improvement
Concordant worsening
Discordant
No change
Open symbols: S4–S1 (supporting)

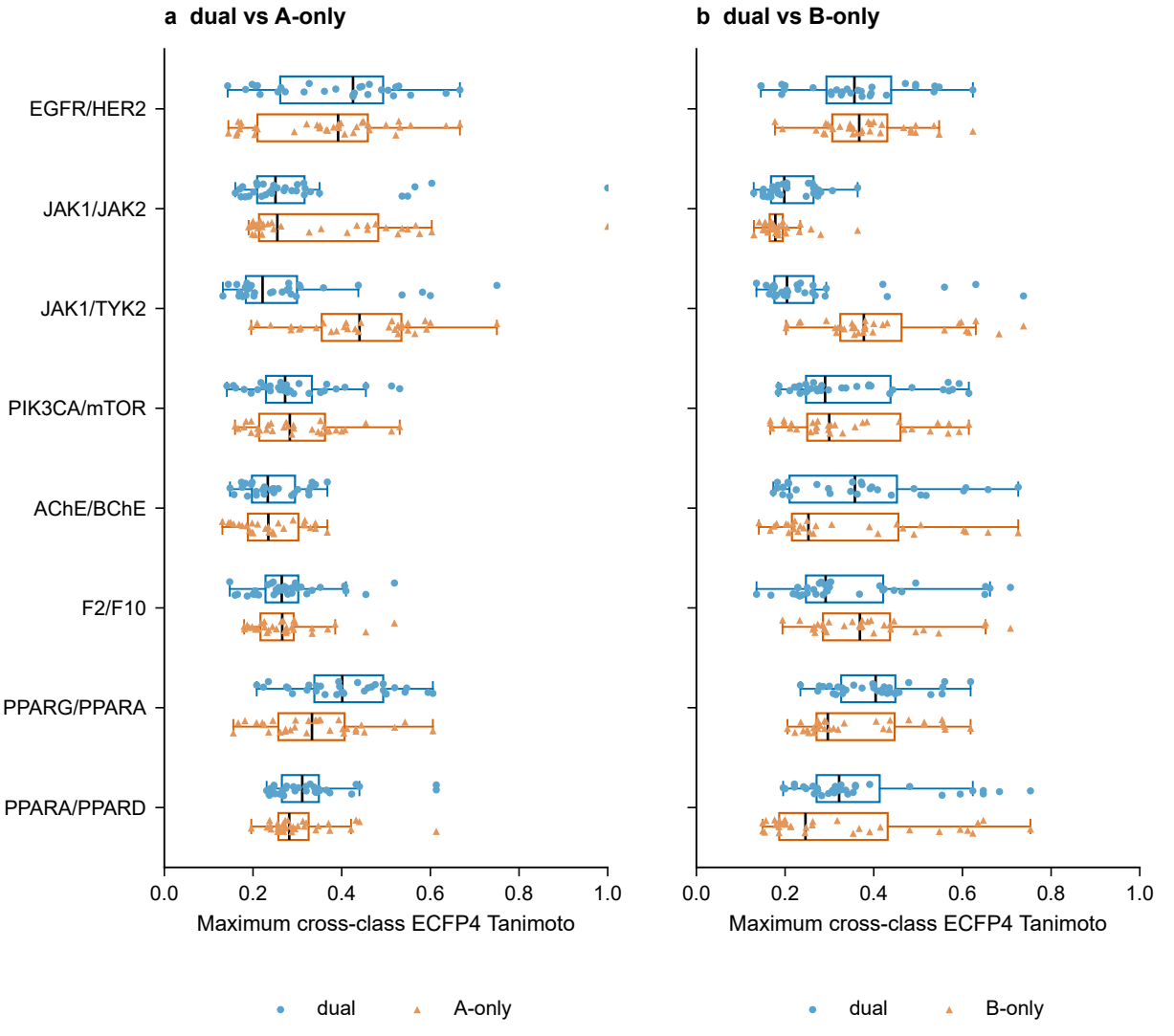

a Top-ranked pose
b Best saved pose
EGFR 3POZ
HER2 3RCD
JAK1 6N7A
JAK2 8BXH
TYK2 3LXP
PIK3CA 4L23
mTOR 4JT6
AChE 4EY7
BChE 4BDS
F2 4UDW
F10 2JKH
PPARG 9V8H
PPARA 6LXA
PPARD 5U3Q
0
2
4
6
8
10
0.0
0.5
1.0
1.5
2.0
2.5
3.0
Heavy-atom RMSD (Å)
Heavy-atom RMSD (Å)
17
29
42
71
101

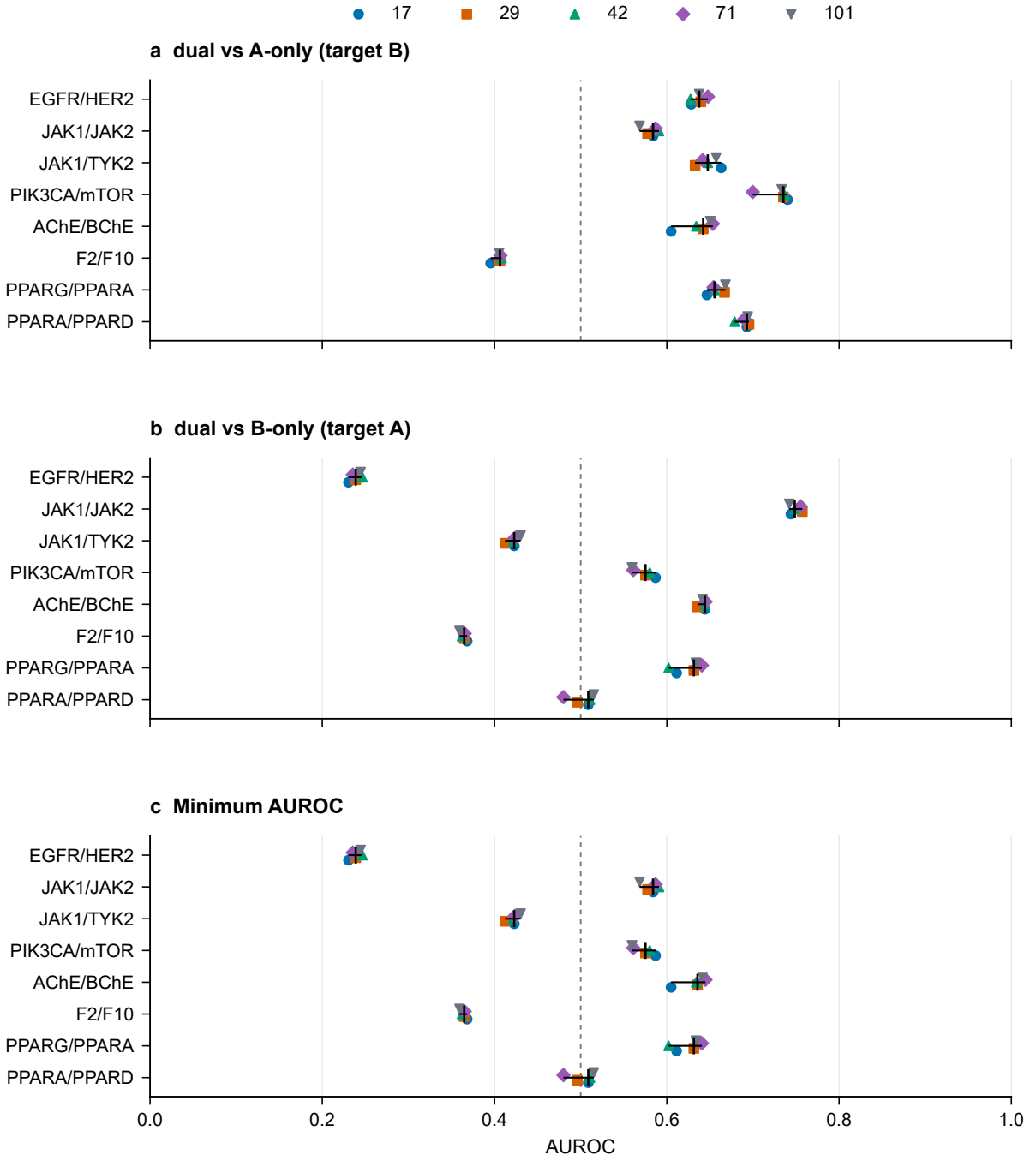

a dual, target A
1 28/28 28/28 28/28 28/28 28/28
2 32/32 32/32 32/32 32/32 32/32
3 32/32 32/32 32/32 32/32 32/32
4 32/32 32/32 32/32 32/32 32/32
5 27/28 27/28 27/28 27/28 27/28
6 32/32 31/32 32/32 31/32 31/32
7 32/32 30/32 32/32 30/32 32/32
8 32/32 27/32 32/32 27/32 32/32
S1 S2 S3 S4 S5
b dual, target B
1 28/28 28/28 28/28 28/28 28/28
2 32/32 32/32 32/32 32/32 32/32
3 32/32 32/32 32/32 32/32 32/32
4 32/32 32/32 32/32 32/32 32/32
5 27/28 27/28 27/28 27/28 27/28
6 32/32 32/32 32/32 32/32 31/32
7 32/32 30/32 32/32 30/32 32/32
8 32/32 27/32 32/32 27/32 32/32
S1 S2 S3 S4 S5
c A-only, target A
1 37/37 37/37 37/37 37/37 37/37
2 32/32 32/32 32/32 32/32 32/32
3 32/32 32/32 32/32 32/32 32/32
4 32/32 32/32 32/32 32/32 32/32
5 25/27 25/27 25/27 25/27 27/27
6 32/32 32/32 32/32 32/32 32/32
7 32/32 32/32 32/32 32/32 32/32
8 32/32 32/32 32/32 32/32 32/32
S1 S2 S3 S4 S5
d A-only, target B
1 37/37 37/37 37/37 37/37 37/37
2 32/32 32/32 32/32 32/32 32/32
3 32/32 32/32 32/32 32/32 32/32
4 32/32 32/32 32/32 32/32 32/32
5 25/27 25/27 25/27 25/27 27/27
6 32/32 32/32 32/32 32/32 32/32
7 32/32 32/32 32/32 32/32 32/32
8 32/32 32/32 32/32 32/32 32/32
S1 S2 S3 S4 S5
e B-only, target A
1 32/32 32/32 32/32 32/32 32/32
2 32/32 32/32 32/32 32/32 32/32
3 32/32 32/32 32/32 32/32 32/32
4 32/32 32/32 32/32 32/32 32/32
5 28/28 27/28 28/28 27/28 28/28
6 32/32 32/32 32/32 32/32 32/32
7 32/32 32/32 32/32 32/32 32/32
8 32/32 32/32 32/32 32/32 32/32
S1 S2 S3 S4 S5
f B-only, target B
1 32/32 32/32 32/32 32/32 32/32
2 32/32 32/32 32/32 32/32 32/32
3 32/32 32/32 32/32 32/32 32/32
4 32/32 32/32 32/32 32/32 32/32
5 28/28 27/28 28/28 27/28 28/28
6 32/32 32/32 32/32 32/32 32/32
7 32/32 32/32 32/32 32/32 32/32
8 32/32 32/32 32/32 32/32 32/32
S1 S2 S3 S4 S5
g neither, target A
1 12/12 12/12 12/12 12/12 12/12
2 14/14 14/14 14/14 14/14 14/14
3 14/14 14/14 14/14 14/14 14/14
4 14/14 14/14 14/14 14/14 14/14
5 15/15 15/15 15/15 15/15 15/15
6 14/14 14/14 14/14 14/14 13/14
7 14/14 14/14 14/14 14/14 14/14
8 14/14 14/14 14/14 14/14 14/14
S1 S2 S3 S4 S5
h neither, target B
1 12/12 12/12 12/12 12/12 12/12
2 14/14 14/14 14/14 14/14 14/14
3 14/14 14/14 14/14 14/14 14/14
4 14/14 14/14 14/14 14/14 14/14
5 15/15 15/15 15/15 15/15 15/15
6 14/14 14/14 14/14 14/14 13/14
7 14/14 14/14 14/14 14/14 14/14
8 14/14 14/14 14/14 14/14 14/14
S1 S2 S3 S4 S5
0 1 2 3 4 5
Unavailable scores (n)

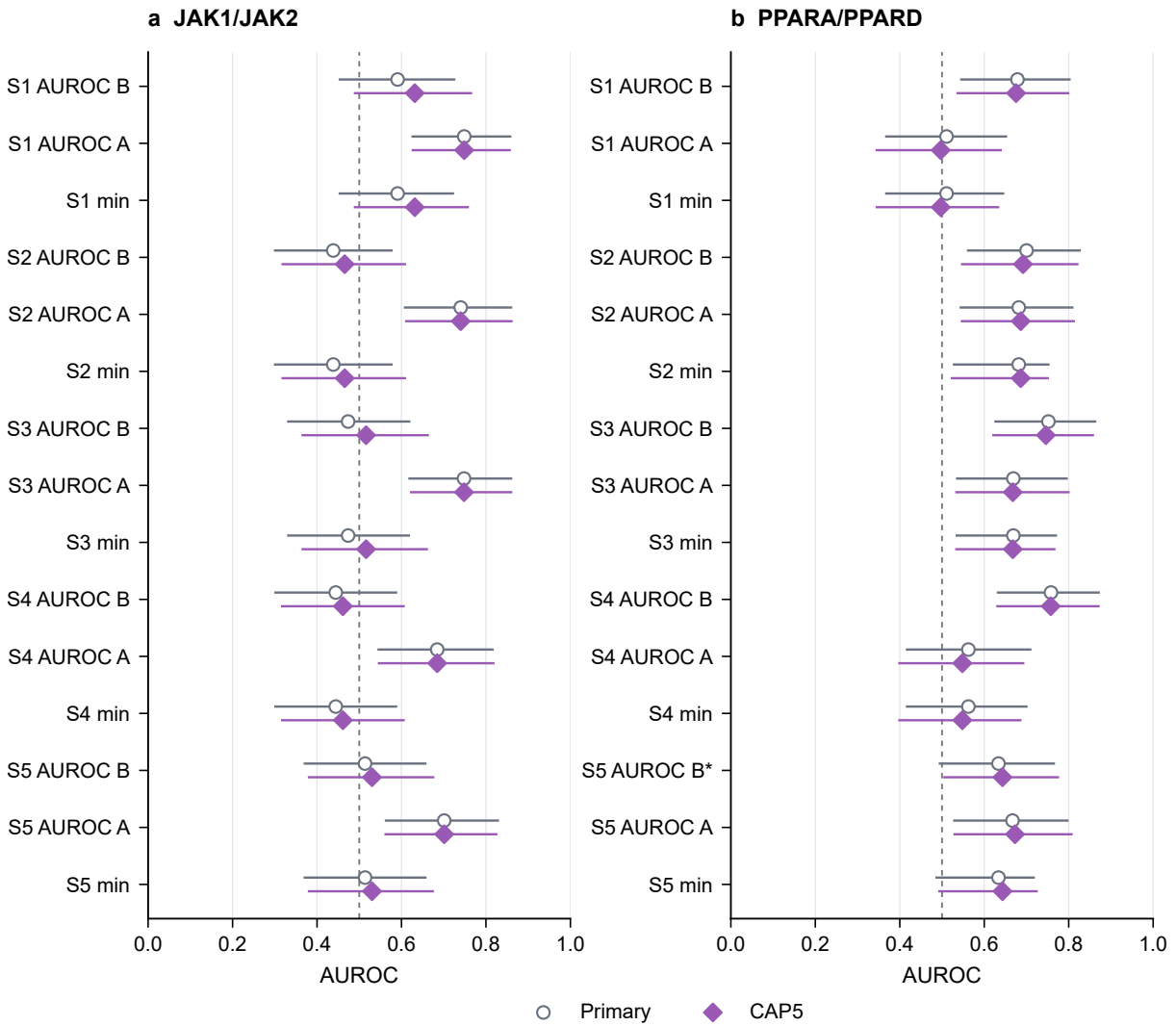

a dual vs A-only
b dual vs B-only
c Minimum AUROC
EGFR/HER2
JAK1/JAK2
JAK1/TYK2
PIK3CA/mTOR
AChE/BChE
F2/F10
PPARG/PPARA
PPARA/PPARD
0.63 0.37 0.52 0.42 0.43
0.59 0.44 0.47 0.44 0.51
0.65 0.66 0.71 0.76 0.67
0.74 0.58 0.70 0.52 0.60
0.63 0.51 0.65 0.54 0.62
0.41 0.54 0.61 0.63 0.56
0.66 0.66 0.65 0.58 0.51
0.68 0.70 0.75 0.76 0.63
0.25 0.22 0.26 0.28 0.25
0.75 0.74 0.75 0.68 0.70
0.42 0.56 0.49 0.68 0.68
0.58 0.66 0.73 0.66 0.71
0.64 0.49 0.49 0.39 0.45
0.36 0.57 0.62 0.57 0.56
0.60 0.28 0.31 0.33 0.45
0.51 0.68 0.67 0.56 0.67
0.25 0.22 0.26 0.28 0.25
0.59 0.44 0.47 0.44 0.51
0.42 0.56 0.49 0.68 0.67
0.58 0.58 0.70 0.52 0.60
0.63 0.49 0.49 0.39 0.45
0.36 0.54 0.61 0.57 0.56
0.60 0.28 0.31 0.33 0.45
0.51 0.68 0.67 0.56 0.63
S1 S2 S3 S4 S5
0.0 0.2 0.4 0.6 0.8 1.0
AUROC


dual
A-only
a Molecular weight (Da)
1000
500
b Calculated LogP
10
5
0
c TPSA (Å²)
200
100
0
d H-bond donors
7.5
5.0
2.5
0.0
e H-bond acceptors
10
5
0
f Rotatable bonds
30
20
10
0
g Ring count
7.5
5.0
2.5
0.0
h Formal charge
2
1
0
−1
1
2
3
4
5
6
7
8

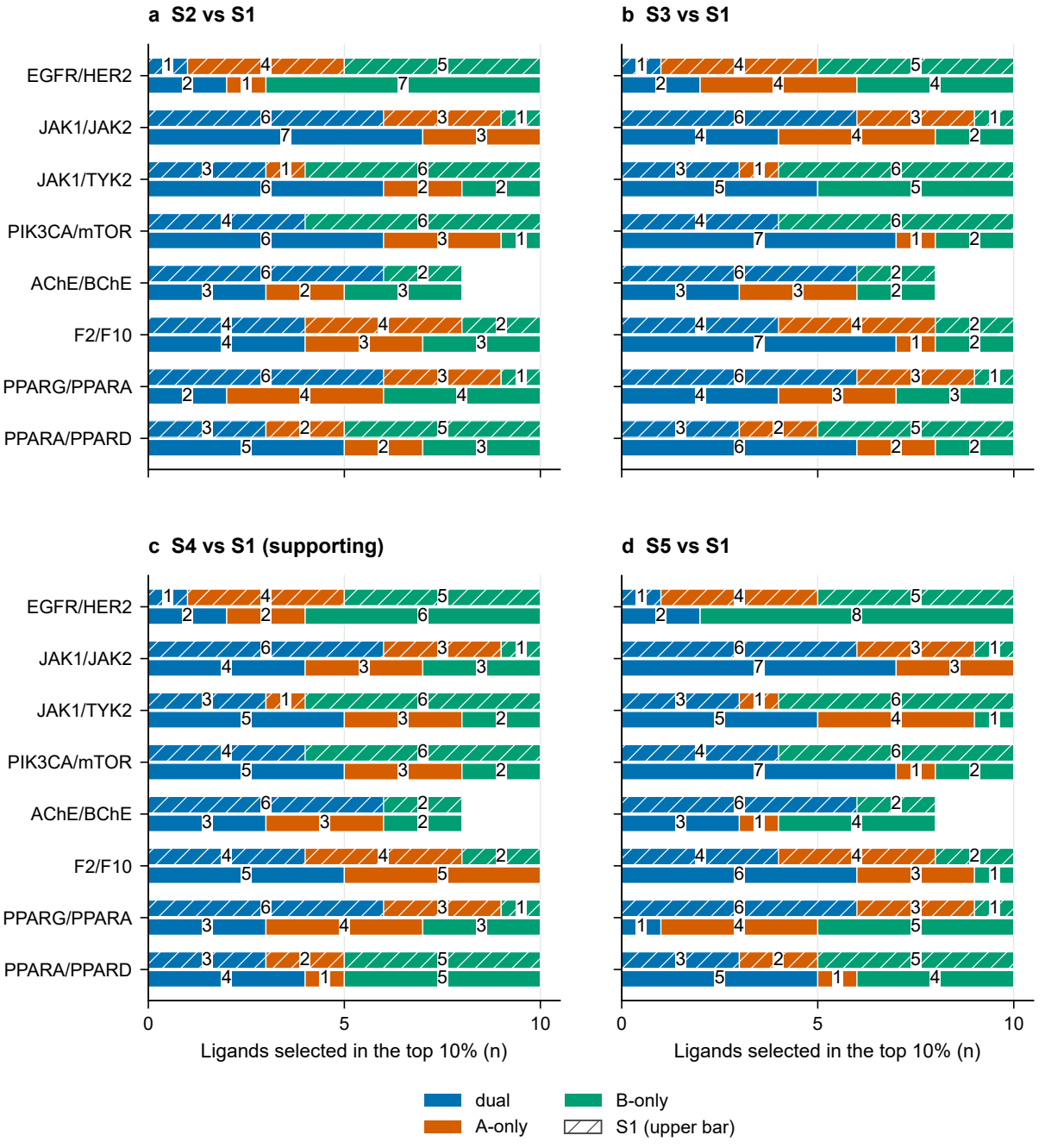

dual
B-only
a Molecular weight (Da)
b Calculated LogP
c TPSA (Å²)
d H-bond donors
e H-bond acceptors
f Rotatable bonds
g Ring count
h Formal charge

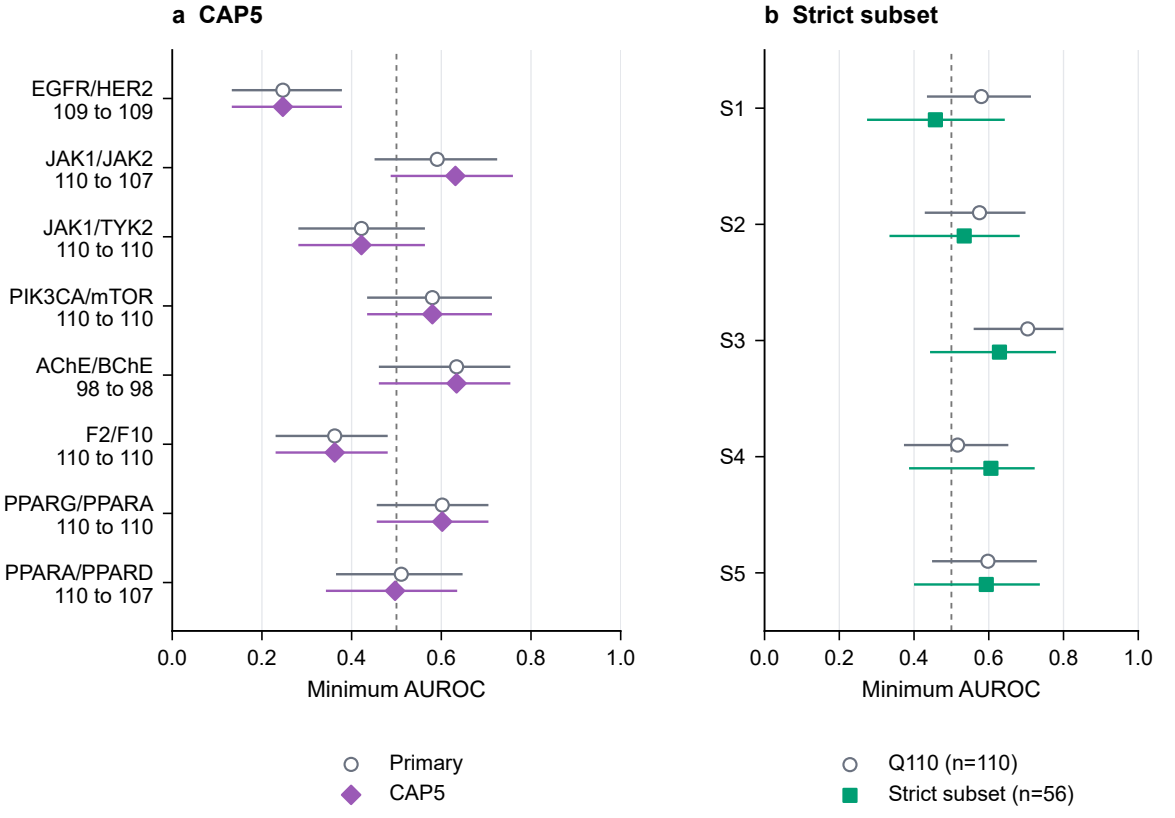

a dual vs A-only
b dual vs B-only
EGFR/HER2
JAK1/JAK2
JAK1/TYK2
PIK3CA/mTOR
AChE/BChE
F2/F10
PPARG/PPARA
PPARA/PPARD
0.0
0.2
0.4
0.6
0.8
1.0
Maximum cross-class ECFP4 Tanimoto
dual
A-only
B-only

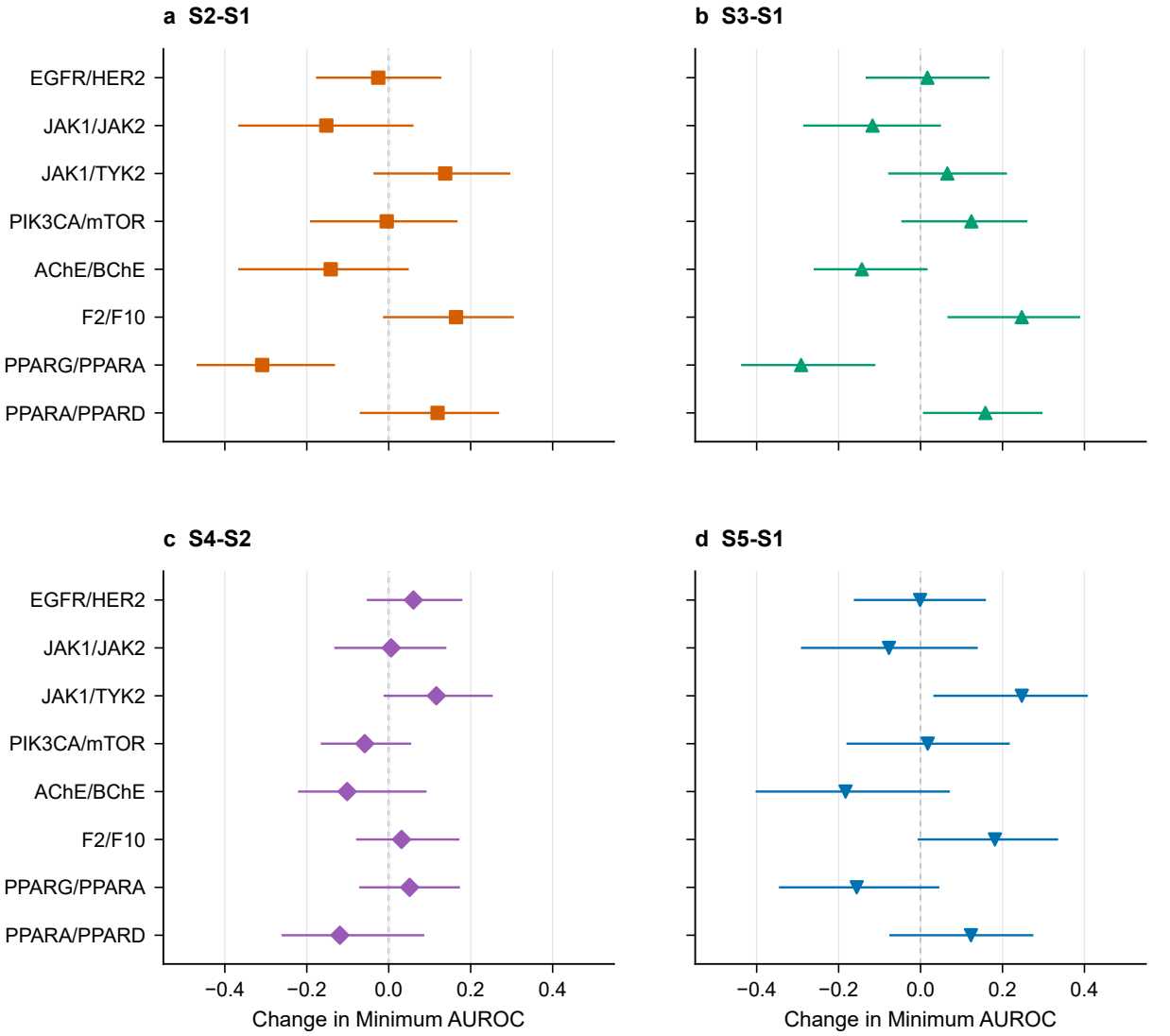

17
29
42
71
101
a dual vs A-only (target B)
EGFR/HER2
JAK1/JAK2
JAK1/TYK2
PIK3CA/mTOR
AChE/BChE
F2/F10
PPARG/PPARA
PPARA/PPARD
b dual vs B-only (target A)
EGFR/HER2
JAK1/JAK2
JAK1/TYK2
PIK3CA/mTOR
AChE/BChE
F2/F10
PPARG/PPARA
PPARA/PPARD
c Minimum AUROC
EGFR/HER2
JAK1/JAK2
JAK1/TYK2
PIK3CA/mTOR
AChE/BChE
F2/F10
PPARG/PPARA
PPARA/PPARD
0.0
0.2
0.4
0.6
0.8
1.0
AUROC

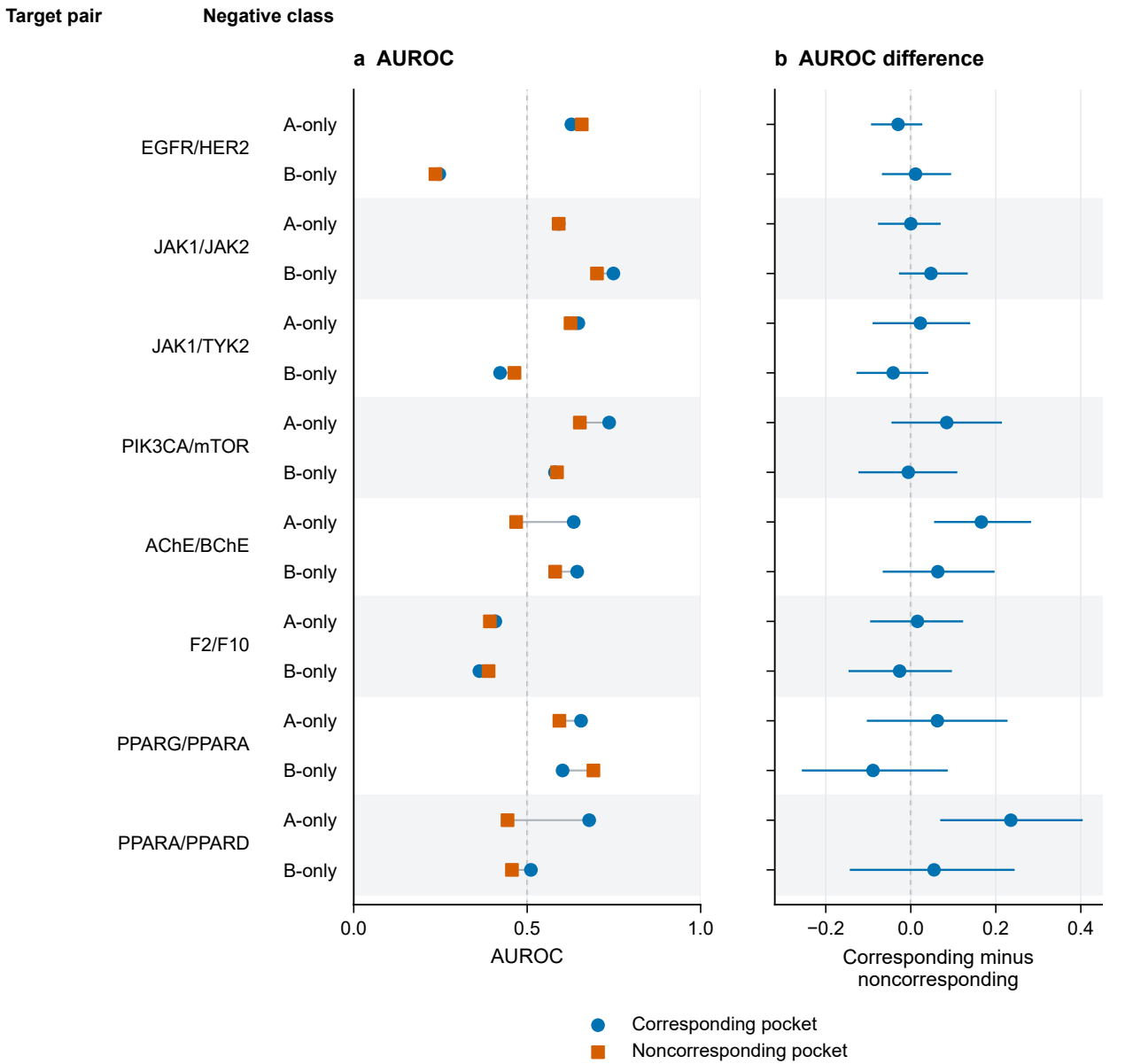


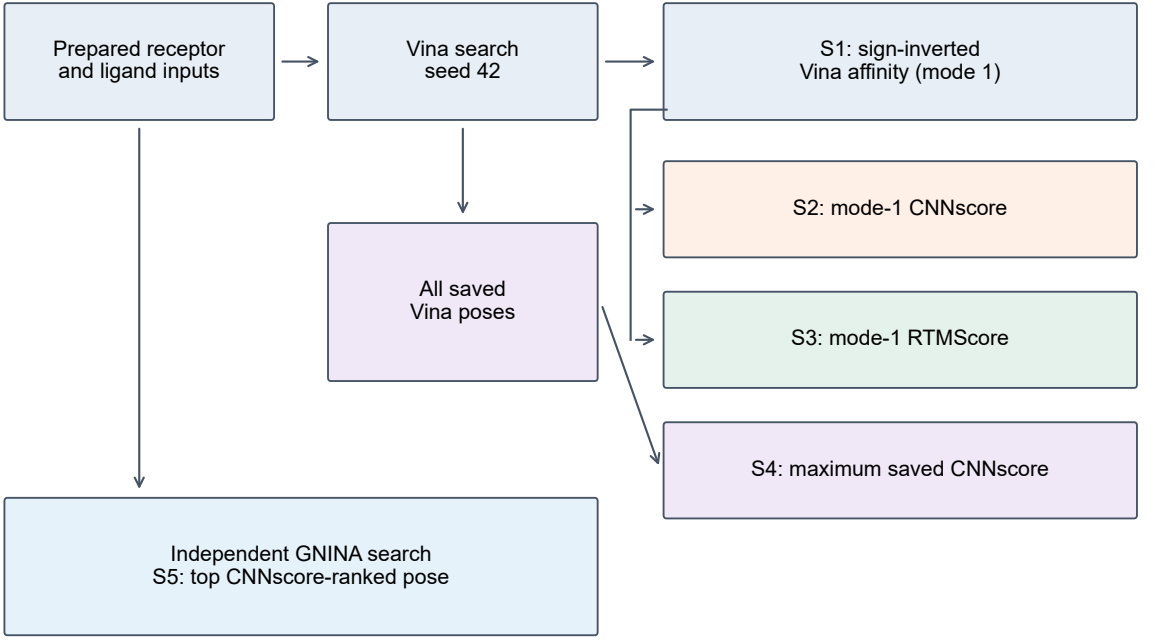
* PPARA/PPARD S5 AUROC B: lower bound of the 95% CI changes from 0.4922 to 0.5030


a S2 vs S1
b S3 vs S1
c S4 vs S1 (supporting)
d S5 vs S1
EGFR/HER2
JAK1/JAK2
JAK1/TYK2
PIK3CA/mTOR
AChE/BChE
F2/F10
PPARG/PPARA
PPARA/PPARD
0
5
10
Ligands selected in the top 10% (n)
dual
A-only
B-only
S1 (upper bar)


a CAP5
EGFR/HER2
109 to 109
JAK1/JAK2
110 to 107
JAK1/TYK2
110 to 110
PIK3CA/mTOR
110 to 110
AChE/BChE
98 to 98
F2/F10
110 to 110
PPARG/PPARA
110 to 110
PPARA/PPARD
110 to 107
0.0
0.2
0.4
0.6
0.8
1.0
Minimum AUROC
Primary
CAP5
b Strict subset
S1
S2
S3
S4
S5
0.0
0.2
0.4
0.6
0.8
1.0
Minimum AUROC
Q110 (n=110)
Strict subset (n=56)


a S2-S1
b S3-S1
c S4-S2
d S5-S1
EGFR/HER2
JAK1/JAK2
JAK1/TYK2
PIK3CA/mTOR
AChE/BChE
F2/F10
PPARG/PPARA
PPARA/PPARD
−0.4
−0.2
0.0
0.2
0.4
Change in Minimum AUROC


Target pair
Negative class
a AUROC
b AUROC difference
EGFR/HER2
JAK1/JAK2
JAK1/TYK2
PIK3CA/mTOR
AChE/BChE
F2/F10
PPARG/PPARA
PPARA/PPARD
A-only
B-only
0.0
0.5
1.0
AUROC
−0.2
0.0
0.2
0.4
Corresponding minus noncorresponding
Corresponding pocket
Noncorresponding pocket


Prepared receptor and ligand inputs
Vina search seed 42
S1: sign-inverted Vina affinity (mode 1)
S2: mode-1 CNNscore
All saved Vina poses
S3: mode-1 RTMScore
S4: maximum saved CNNscore
Independent GNINA search
S5: top CNNscore-ranked pose